UNIT III: Classification and Regression

Q1. What are Decision Trees? Explain with the help of example.

A **Decision Tree** is a supervised machine learning algorithm used for both **classification** and **regression** tasks. It models decisions and their possible consequences as a tree-like graph of nodes and branches.

Each **internal node** represents a feature (attribute), each **branch** represents a decision (rule), and each **leaf node** represents an outcome (label/class).

Key Concepts:

- **Root Node:** First node representing the entire dataset.
- **Splitting:** Dividing dataset into subsets based on feature values.
- **Leaf/Terminal Node:** Node that contains the final outcome (class label).
- **Entropy:** Measure of disorder or impurity.
- **Information Gain:** Measures reduction in entropy. Used to decide which feature to split on.

How it Works:

1. Select the best feature to split the dataset (using Information Gain, Gini Index, etc.).
2. Split the dataset into subsets.
3. Repeat the process recursively for each child until one of the stopping conditions is met:
   - All instances belong to one class.
   - No more features to split.
   - A predefined depth is reached.

Example: Let's say we want to predict if a person will buy a laptop based on their **Age** and **Income**.

Age	Income	Buy Laptop
< 30	High	No
< 30	Medium	Yes
31–40	High	Yes
> 40	Medium	No
> 40	Low	No
31–40	Medium	Yes

A simplified **decision tree** could be:

```
            Age
          /  |  \
      <30  31-40  >40
      /       \     \
 Income      Buy=Yes  Buy=No
  /  \
High  Med
No    Yes
```

This tree can now be used to classify new individuals based on their age and income.

Advantages: Easy to understand and visualize**,** Requires little data preprocessing**,** Can handle both numerical and categorical data

Disadvantages: Prone to overfitting**,** Unstable to slight data variations**,** Can be biased if some classes dominate

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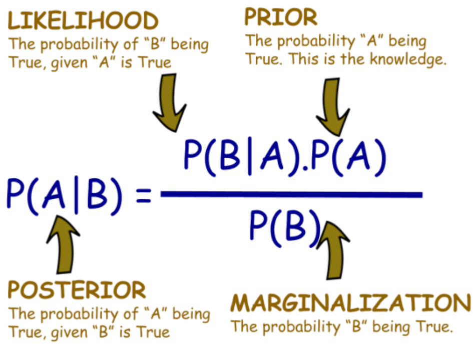
Q2. What is Naïve Bayesian Classifier? Explain with the help of Example.

The **Naïve Bayes Classifier** is a **probabilistic supervised learning algorithm** based on **Bayes' Theorem**. It assumes that all features are **independent** of each other given the class label — hence the term *"naïve."*

It is particularly used for **classification** tasks like spam detection, sentiment analysis, and document categorization.

Bayes' Theorem:



How it Works:

1. Calculate prior probabilities for each class.
2. Compute likelihood of each feature value given the class.
3. Apply Bayes' theorem to calculate posterior probability.
4. Choose the class with the highest posterior probability.

Example: Email Spam Classification, Let's classify whether an email is **Spam** or **Not Spam** using two keywords: **"Buy"** and **"Discount"**.

Email	**Buy**	**Discount**	**Class**
E1	Yes	Yes	Spam
E2	Yes	No	Spam
E3	No	Yes	Not Spam
E4	No	No	Not Spam

We want to classify a **new email** that contains **"Buy = Yes"** and **"Discount = Yes"**.

Step-by-step:

1. Prior Probabilities:
   - P(Spam) = 2 / 4 = 0.5
   - P(Not Spam) = 2 / 4 = 0.5
2. Likelihoods:
   - P(Buy=Yes | Spam) = 2 / 2 = 1
   - P(Discount=Yes | Spam) = 1 / 2 = 0.5
   - P(Buy=Yes | Not Spam) = 0
   - P(Discount=Yes | Not Spam) = 1 / 2 = 0.5
3. Posterior Probabilities (ignoring P(X)):
   - P(Spam | X) $\propto$ 1 x 0.5 x 0.5 = 0.25
   - P(Not Spam | X) $\propto$ 0 x 0.5 x 0.5 = 0

Hence, the email is classified as **Spam**.

Advantages: Simple and fast**,** Performs well with high-dimensional data (e.g., text classification)**,** Works well even with small training datasets

Disadvantages: Assumes independence among features (which is rarely true in real life)**,** Can't learn complex relationships between features

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Q3. Where Regression Methods Are Applied? How It Plays Important Role in Data Analysis?

Regression is a type of **supervised learning** used to model and analyze the relationship between a **dependent variable (target)** and one or more **independent variables (predictors)**. The goal is to **predict continuous numeric values**.

Applications of Regression Methods:

Field	Application Example
Business	Predicting sales, customer lifetime value, demand forecasting
Finance	Stock price prediction, risk analysis
Healthcare	Estimating disease progression, dosage recommendation
Real Estate	Predicting property prices based on location, size, etc.
Marketing	Customer behavior modeling, advertisement ROI prediction
Environmental Science	Forecasting temperature, pollution levels
Agriculture	Yield prediction, soil moisture estimation
Sports	Predicting player performance, team rankings

Importance of Regression in Data Analysis:

1. **Predictive Modeling:** Regression provides a mathematical model that can be used to make predictions about future outcomes.
2. **Trend Analysis:** Helps identify trends over time, such as seasonal demand or economic cycles.
3. **Risk Assessment:** Useful in determining factors contributing to risks, such as in insurance or credit scoring.
4. **Decision Making:** Regression models offer quantitative support for business decisions by estimating the impact of different variables.
5. **Optimization:** Used to identify which input features should be increased or reduced to improve the outcome.

Types of Regression Models:

- **Linear Regression** – Predicts a linear relationship between variables.
- **Multiple Linear Regression** – Involves more than one independent variable.
- **Polynomial Regression** – Models non-linear relationships.
- **Logistic Regression** – Used for binary classification (even though it's called regression).
- **Ridge, Lasso, ElasticNet** – Regularized regression techniques to avoid overfitting.

Q4. Explain Working of Nearest Neighbor Classifier with Suitable Example

K-Nearest Neighbors (KNN) is a **supervised classification algorithm** that classifies a new data point based on the **majority class among its k nearest neighbors** in the feature space.

Working of KNN Algorithm:

1. **Choose a value of K** (number of neighbors to consider).
2. **Calculate the distance** (e.g., Euclidean, Manhattan) between the new data point and all existing data points in the dataset.
3. **Sort all distances** and find the **K closest neighbors**.
4. **Assign the class label** based on the **majority vote** from those K neighbors.

Example: Suppose you have a dataset of fruits with the following features:

Fruit	Weight (g)	Size (cm)	Label
Apple	180	8	Apple
Apple	200	9	Apple
Orange	160	7.5	Orange
Orange	170	7.8	Orange

Now, classify a new fruit with weight = 175g and size = 8cm using K=3.

1. **Calculate distance** from the new point to all existing data points using Euclidean distance:

$$d = \sqrt{(x_1 - x_2)^2 + (y_1 + y_2)^2}$$

2. **Sort distances** and select the 3 nearest neighbors.
3. Among the nearest neighbors, suppose 2 are Apples and 1 is Orange.
4. Therefore, **the new fruit is classified as Apple** (majority vote).

Advantages of KNN: Simple to understand and implement. No training phase (lazy learning). Works well with small to medium-sized datasets.
Disadvantages: Computationally expensive during prediction. Performance degrades with high dimensionality (curse of dimensionality). Sensitive to irrelevant features and noise.
Use Cases: Recommender systems**,** Handwriting recognition**,** Medical diagnosis**,** Image classification

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Q5. Explain the Different Types of Classification

Classification is a **supervised learning** technique used to categorize data into predefined classes or labels. Based on various criteria, classification problems can be divided into different types:

Type of Classification	Description	Example	Extra
1. Binary Classification	Classifies data into **two classes**.	Spam vs. Not Spam, Disease vs. No Disease	Algorithms: Logistic Regression, SVM(binary), Decision Trees
2. Multi-class Classification	Classifies data into **more than two classes**.	Handwritten digit recognition (0–9), Weather types (Sunny, Rainy, Cloudy)	Algorithms: Random Forest, Naive Bayes, Neural Networks.
3. Multi-label Classification	Each instance can belong to **multiple classes simultaneously**.	A news article can be labeled as "Politics", "Economy", and "World"	Output is a set of labels. Common in text classification, image tagging
4. Imbalanced Classification	One class has **significantly more instances** than the others. Needs resampling, cost-sensitive learning, or anomaly detection techniques.	Fraud detection (legit transactions >> fraud transactions)	Evaluation metrics like Precision, Recall, F1-score are more relevant than accuracy.
5. Ordinal Classification	Classes have a **natural order**, but not a numeric difference.	Movie rating: Poor, Average, Good, Excellent	Useful in surveys or satisfaction ratings

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Q6. Given a real-world scenario, how would you use the KNN algorithm to solve a practical problem?

Scenario: Predicting Whether a Patient Has Diabetes: Suppose a healthcare organization wants to predict whether a patient has diabetes based on health metrics like glucose level, BMI, blood pressure, and age. This is a **binary classification** problem (Diabetic / Not Diabetic).

Steps to Solve Using K-Nearest Neighbors (KNN):

Step	Description
1. **Collect Data**	Gather a dataset containing patient records with known labels (diabetic or not). E.g., Pima Indian dataset.
2. **Preprocess Data**	Handle missing values, normalize data (e.g., scale glucose, BMI).
3. **Split Dataset**	Divide into training and testing sets (e.g., 80% training, 20% testing).
4. **Choose Value of K**	Try different K values (e.g., K=3, 5, 7) and select the one giving best accuracy on validation set.
5. **Compute Distances**	Use Euclidean distance to find the closest K training samples to a new test point.
6. **Predict Label**	Based on majority class of K nearest neighbors (e.g., if 3 neighbors: [Yes, No, Yes] → Predict: Yes).
7. **Evaluate Model**	Use accuracy, precision, recall, or F1-score to evaluate performance on the test set.

Advantages in This Use Case: Simple, interpretable model. No assumptions about data distribution. Effective when the dataset is small and well-labeled.

Limitations: Slow for large datasets. Sensitive to irrelevant features and imbalanced data.

Conclusion: KNN provides a straightforward way to classify new patients based on similarity to known cases. It's especially effective when class boundaries are complex and not easily captured by linear models.

Q7. Differentiate between Decision Tree and Random Forest

Criteria	Decision Tree	Random Forest
Definition	A flowchart-like structure where each internal node represents a test on a feature	An ensemble of multiple decision trees combined to produce better accuracy
Model Type	Single model (tree)	Ensemble model (collection of trees)
Training	Grows one tree based on entire dataset	Trains multiple trees on random subsets (bagging) and features
Overfitting	Prone to overfitting, especially on noisy data	Reduces overfitting through averaging
Accuracy	Generally less accurate	Higher accuracy due to aggregation of multiple trees
Interpretability	Easy to interpret and visualize	Less interpretable due to many trees
Training Speed	Faster to train	Slower due to training multiple trees
Prediction Speed	Faster (single tree traversal)	Slower (multiple trees need to be evaluated)
Robustness	Less robust to data fluctuations	More robust and stable due to ensemble nature
Use Case Example	Quick prototype modeling, small datasets	Complex classification/regression, larger datasets

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Q8. Explain different types of regression

Regression is a supervised learning technique used to predict **continuous numerical values**. There are several types of regression models based on the nature of the relationship between dependent and independent variables.

1. Linear Regression

Feature	Description
Model Type	Linear
Formula	y=a+b1x1+b2x2+...+bnxny = a + b_1x_1 + b_2x_2 + ... + b_nx_n
Use Case	Predicting house prices, sales forecasting
Assumption	Linear relationship between independent and dependent variables

2. Multiple Linear Regression

Feature	Description
Model Type	Linear (but with more than one input variable)
Formula	Same as linear regression, but with multiple variables
Use Case	Predicting salary based on education, experience, and age

3. Polynomial Regression

Feature	Description
Model Type	Non-linear
Formula	y=a+b1x+b2x2+b3x3+...+bnxny = a + b_1x + b_2x^2 + b_3x^3 + ... + b_nx^n
Use Case	Modeling curved relationships, e.g., growth curves
Advantage	Captures non-linearity better than linear regression

4. Ridge Regression

Feature	Description
Model Type	Regularized linear regression
Use Case	When multicollinearity exists in data
Advantage	Prevents overfitting by adding L2 penalty

5. Lasso Regression

Feature	Description
Model Type	Regularized linear regression
Use Case	Useful for feature selection
Advantage	Adds L1 penalty and can shrink some coefficients to zero

6. Logistic Regression ***(Though classification, it's called regression)***

Feature	Description
Model Type	Classification (binary or multi-class)
Use Case	Spam detection, disease prediction (Yes/No)
Output	Probability (0 to 1), mapped using the logistic (sigmoid) function

7. Support Vector Regression (SVR)

Feature	Description
Model Type	Extension of Support Vector Machines for regression
Use Case	Works well on non-linear data with high dimensionality
Advantage	Robust to outliers

8. Decision Tree Regression

Feature	Description
Model Type	Non-linear, rule-based
Use Case	Can handle both linear and non-linear data effectively
Advantage	No need for feature scaling
